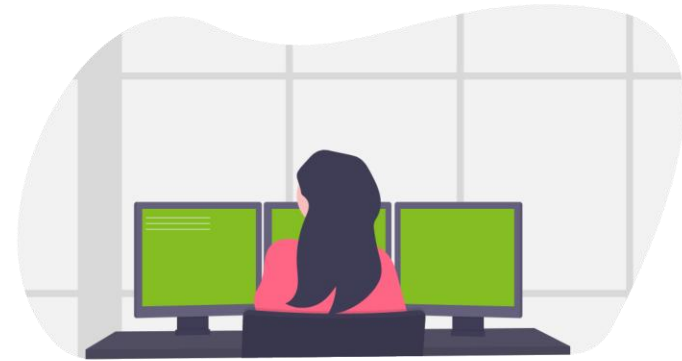
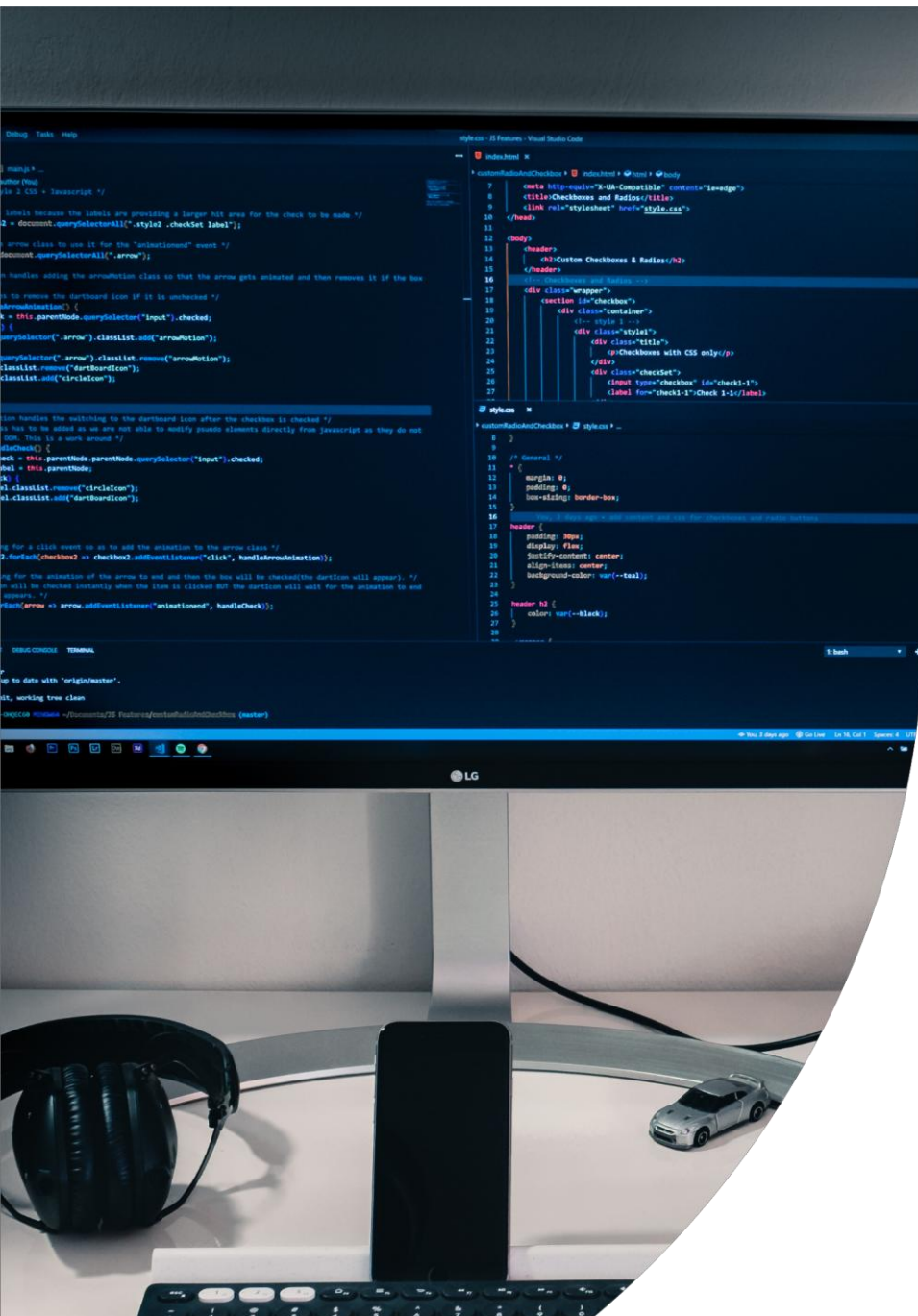


Software Development Proposal



https://github.com/adonisenriquez-lab/Daikibo_Telemetry

Software Development Proposal

Deloitte.

1. Overview

PROBLEM

Daikibo receives industrial telemetry in at least two incompatible formats (Format A and Format B) from multiple plants. This fragmentation prevents: Obtaining a unified view of the Shop Floor by machine, plant, and timestamp. Calculating consistent OEE metrics across equipment. Converting technical signals into automated business actions for SAP. The current Excel-based approach does not scale for high-frequency telemetry and creates uncontrolled cost risks in SAP CO.

PROPOSAL

A two-layer analytical pipeline normalizes, classifies, and consolidates IIoT telemetry into a structured data contract: Analytical Layer: standardizes formats, validates contracts, and calculates OEE metrics (starting with Availability) Transactional Layer: sends only processed and actionable events to SAP PM (preventive maintenance), SAP MM (JIT spare parts), and SAP CO (cost variance analysis) The PaaS/serverless architecture (AWS Lambda, Databricks) ensures scalability without idle infrastructure costs.

NEXT STEPS

To validate the business case assumptions before committing enterprise investment, a 3-sprint, 6-week pilot is proposed (USD 35,000) with a formal GO/NO-GO decision at the close of Sprint 3, based on: technical pipeline feasibility, compatibility with real Daikibo data, operational utility of alerts, and confirmation of the savings baseline.

USD 612,500

Estimated annual savings

50 critical machines in scope,
70% of failures convertible to
preventive.

2.4 months

Estimated payback

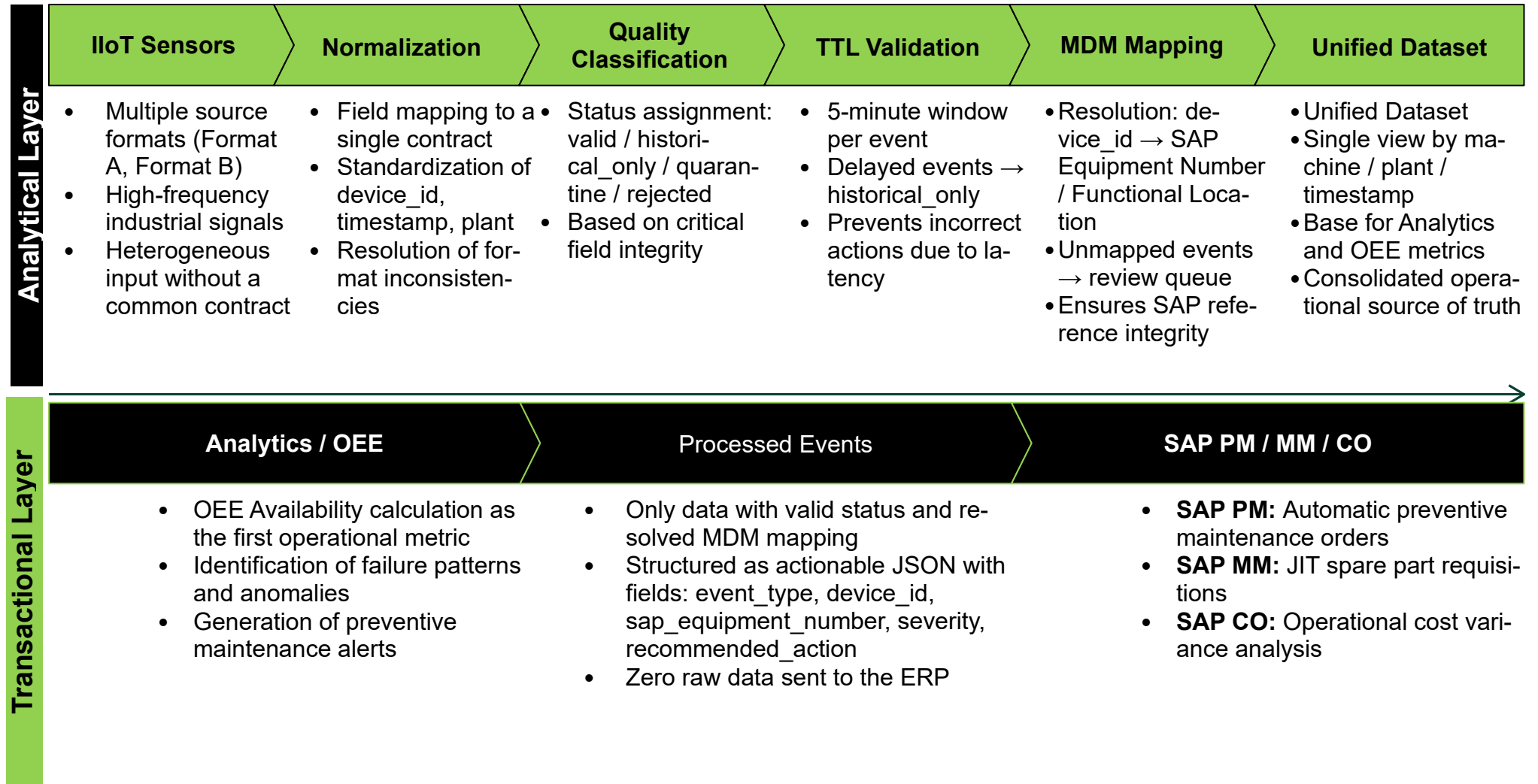
Base case with positive ROI
across all sensitivity scenarios

6 weeks

validation pilot

3 sprints ·
USD 35,000 ·
GO/NO-GO decision at closing

2. Scope



Software Development Proposal

✓  Daikibo Factory Meiyo	Last update: <1min ago ◀
✓  Daikibo Factory Seiko	Last update: <1min ago ◀
✓  Daikibo Berlin	Last update: <1min ago ◀
✗  Daikibo Shenzhen	Last update: <1min ago ▾
✗  CNC	Last update: 2min ago ▾
✗  Status: Unhealthy	2min ago
✓  Status: Healthy	12min ago
Load More	
✓  LaserCutter	Last update: <1min ago ◀
✓  HeavyDutyDrill	Last update: <1min ago ◀
✓  SpotWelder	Last update: <1min ago ◀
✓  LaserWelder	Last update: <1min ago ◀
✓  MetalPress	Last update: <1min ago ◀
✓  Furnace	Last update: <1min ago ◀
✓  ConveyorBelt	Last update: <1min ago ◀
✓  AirWrench	Last update: <1min ago ◀

3. Estimate

Sensitivity table: scenario analysis

Scenario	Failures Converted / Year	Estimated Annual Savings	Investment	ROI
Optimist	35 failures	USD 1,225,000	USD 250,000	4.9x
Base	17-18 failures	USD 612,500	USD 250,000	2.5x
Pessimistic	7-8 failures	USD 262,500	USD 250,000	1.1x

Business case assumptions

- 50 critical machines in scope
- 1 critical failure per machine / year
- Corrective cost: USD 50,000 per failure
- Preventive cost: USD 15,000 per intervention 70% of failures convertible to preventive
- Scaling investment: USD 250,000



This simulation is illustrative. The enterprise investment decision depends on the validation with real data obtained during the 6-week pilot.

4. Timeline

Active Quick

ACTIVITY	COST	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Sprint 1: UCI Hydraulic Validation	USD 4,200						
Ingestion and normalization of UCI Hydraulic dataset	—		2,205 cycles · >95% classified · <2 sec/cycle				
Validation of data contract and OEE metrics	—						
Sprint 2: Validation with a real	USD 11,000						
Connection to 1 real machine and MDM mapping resolution	—				>90% classified events · master mapping available		
Identification of format gaps and pipeline iteration	—						
Sprint 3: Alert validation with a	USD 14,000						
Generation and review of maintenance alerts	—						False positives <20% · ≥ 1 anomaly validated by an expert
GO/NO-GO decision with CFO and CEO	—						
Expected Quick Wins	—		Reproducible pipeline (CTO/IT)		Normalized Daikibo data (Plant/Ops)		Validated useful alert (CFO/COO)

Total pilot investment: USD 35,000 (incl. 20% contingency) · GO/NO-GO decision at the close of Sprint 3 based on real operational evidence before committing USD 250,000